Unified Image Deraining via Degradation Guidance and Semantic Restoration

Xieqian Ying, Chunxiao Liu

***Abstract*—Unified image deraining aims to handle diverse rain types with a single model, but the heterogeneity of different rain types makes it challenging to achieve effective rain removal while preserving background details. This paper presents a unified image deraining framework via degradation guidance and semantic restoration. To provide degradation guidance, a mixture-of-experts degradation evaluator extracts degradation information of rainy images from multiple perspectives and generates compact degradation representations to construct the state-space parameters in the restoration network. For semantic restoration, a retrained DINOv3 can recover clean semantic representations from rainy images to modulate the decoder features in the restoration network, improving rain-background discrimination and detail reconstruction. Furthermore, a three-stage training strategy is introduced to bridge the gap between clean-image assistance and rainy-image-only inference. First, a teacher degradation evaluator learns reference degradation representations through paired rainy-clean images, while a frozen pretrained DINOv3 provides clean semantic representations through clean images. Second, another DINOv3 is retrained to reproduce these clean semantic representations directly from rainy images. Third, a student degradation evaluator learns to reproduce the reference representations from rainy images alone while jointly optimized with DINOv3 and the restoration network. Extensive experiments on the RainRAG benchmark and real-world samples demonstrate the superior unified deraining performance and strong generalization ability of the proposed method. The code is released at https://github.com/Gin-yxq/UniDR_DINO.**

***Index Terms*—Unified image deraining, degradation guidance, semantic restoration.**

I. Introduction

IMAGE deraining aims to remove rain-induced degradation and recover clear background content. As a fundamental image restoration task, it has been applied in various outdoor vision scenarios, including security surveillance, autonomous driving, film production, and live sports broadcasting. In real-world environments, the visual characteristics of rain vary considerably with both rain form and illumination conditions. During the daytime, rain streaks usually appear as thin, elongated, and semitransparent structures, whereas raindrops tend to appear as irregular droplets or water traces that cause refraction and blurring. At night, the visibility and appearance of rain streaks become highly dependent on local artificial illumination, and raindrops interacting with light sources can produce glare and artifacts.

Traditional image deraining methods primarily relied on conventional digital image processing techniques [1]–[3]. By manually designing physical models or incorporating prior knowledge, these methods were able to remove rain to some extent, but generally suffered from limited performance and were mostly designed for daytime rain streaks. With the emergence of deep learning, data-driven approaches were introduced to image deraining, where neural networks were employed to learn the mapping from rainy images to clean images, leading to significant advances in image deraining performance. However, existing image deraining datasets [4]–[15] typically contain images with only a single rain type in both the training and test sets. Consequently, most deep learning-based image deraining methods are developed as single-task models [16]–[36], targeting a specific rain type or requiring separate model weights for different rain types. In practical applications, single-task models designed for only one rain type cannot simultaneously cope with diverse and complex rainy conditions. Although separate model weights can be trained for different rain types, it remains difficult to adaptively switch between models according to specific scenarios. Motivated by these limitations, recent studies have introduced the concept of unified image deraining [37]–[39], aiming to employ a single set of model weights to handle multiple rain types, along with the construction of corresponding unified image deraining datasets.

Compared with single-task deraining, unified image deraining faces several unique challenges. First, different rain types introduce highly diverse degradation patterns, with substantial variations in rain scale, density, and geometric appearance, making it difficult for a single network to comprehensively model such complex degradation representations. Second, training samples from different rain types may require significantly different parameter update directions, which can easily lead to gradient conflicts during optimization. Moreover, when rain exhibits large scale variations or shares similar patterns with background structures, the model may suffer from rain-background confusion, resulting in incomplete rain removal, unintended removal of background details, and difficulty in recovering degraded background information. However, existing unified image deraining methods have not fully addressed

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these challenges, so their performance remains limited.

To address the above challenges, a unified image deraining framework based on degradation guidance and semantic restoration is proposed. The framework consists of a mixture-of-experts degradation evaluator, a state-space-based restoration network, and a retrained DINOv3 [40]. The mixture-of-experts degradation evaluator estimates the degradation characteristics of rainy images from multiple perspectives and generates compact 1D and 2D degradation representations, which implicitly model the overall type and spatial distribution of rain. These representations are then introduced into the restoration network to provide degradation-aware guidance. In particular, the 1D degradation representation is utilized to condition the feature transformations, while the 2D degradation representation is employed to dynamically construct key state-space parameters, enabling the restoration process to adapt to different rain degradations and mitigating optimization conflicts caused by their heterogeneity. In parallel, DINOv3 is retrained to recover clean semantic representations directly from rainy images. The recovered semantic representations are incorporated into the decoder of the restoration network through a semantic compensation module, providing additional semantic cues for distinguishing rain from background structures and restoring degraded details.

To make both degradation guidance and semantic restoration available during inference without access to clean images, a three-stage teacher–student training strategy is proposed. In the first stage, a teacher degradation evaluator is trained jointly with the restoration network to learn reference degradation representations from rainy and clean image pairs, while a frozen pretrained DINOv3 model extracts clean semantic representations from clean images. In the second stage, a student DINOv3 is retrained to recover clean semantic representations directly from rainy images. In the third stage, a student degradation evaluator is trained to reproduce the reference degradation representations using rainy images alone, and the student degradation evaluator, DINOv3, and the restoration network are jointly optimized to obtain the final unified deraining model.

Fig. 1 presents the comparison of different image deraining paradigms. The main contributions of this work are summarized as follows:

- A degradation guidance mechanism is proposed to adapt the restoration process to heterogeneous rain types, allowing a single model to perform a degradation-informed and input-adaptive restoration.
- A semantic restoration mechanism is introduced to improve rain-background discrimination and detail reconstruction. Clean semantic representations are recovered from rainy images and incorporated into the restoration process to provide complementary semantic cues.
- A three-stage teacher–student training strategy is developed to transfer the knowledge learned under clean-image supervision while enabling rainy-image-only inference. The strategy progressively transfers degradation and semantic representations learned with clean-image supervision to the final unified deraining model.

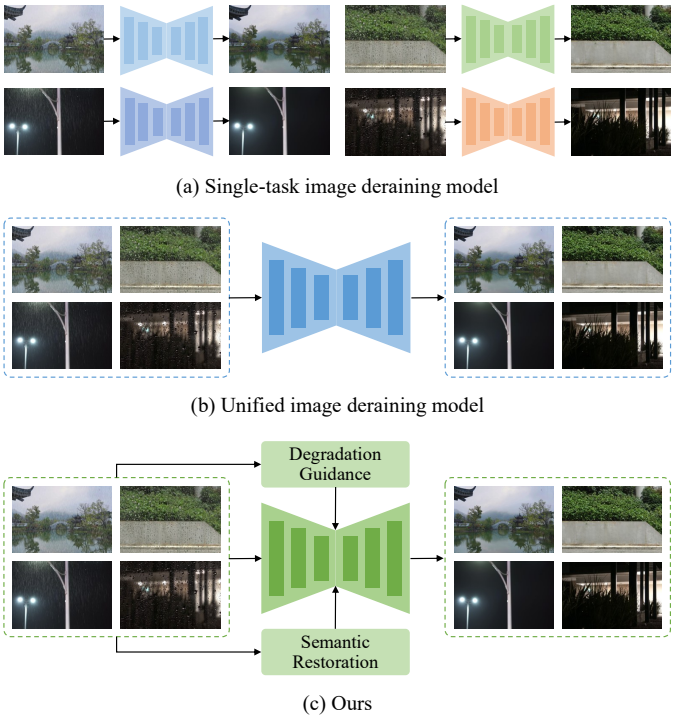


Fig. 1. Comparison of different image deraining paradigms.

II. Related Work

A. Traditional Image Deraining Methods

Traditional image deraining methods are rooted in early digital image processing techniques, utilizing hand-crafted physical models or prior knowledge to achieve rain removal. Luo et al. [1] formulated single-image deraining using a nonlinear screen blend model, and employed discriminative sparse coding over a learned dictionary with a mutual exclusivity constraint to jointly separate the rain and derained image layers. Li et al. [2] proposed a layer decomposition framework with patch-based priors, where Gaussian mixture models are learned separately for background and rain layers, together with gradient sparsity regularization to guide the separation of rain streaks from image content. Zhu et al. [3] proposed a joint bi-layer optimization framework for single-image deraining, which estimates the dominant rain direction from local gradient statistics and exploits centralized sparse representation, directional constraints, and rain-patch similarity priors to jointly separate background and rain layers. However, these methods are primarily designed for daytime rain streaks and often fail when test samples deviate from the assumptions underlying the adopted physical models or priors.

B. Single-task Image Deraining Models

Deep learning-based image deraining methods utilize neural networks to learn the complex mapping relationship from rainy images to clear images, initially employing a convolutional neural network (CNN) architecture. Fan et al. [16] proposed a residual-guided feature fusion network, which adopts a cascaded architecture where residuals from shallower blocks guide deeper blocks for coarse-to-fine rain removal, with

recursive convolution and intermediate supervision that further enhances progressive reconstruction. Ren et al. [17] proposed a progressive recurrent network, which recursively unfolds a shallow ResNet into multiple stages and introduces recurrent connections to exploit feature dependencies across stages for progressively removing rain streaks. Deng et al. [18] proposed a detail-recovery deraining network that separates rain removal and detail restoration into two parallel sub-networks, using SE-enhanced [41] residual blocks to remove rain streaks and a structure detail context aggregation block to recover the details lost during deraining. Yi et al. [19] exploited the residue channel prior [42] extracted from rainy images as structural guidance for deraining, and introduced wavelet-based multi-level learning, interactive feature fusion, and iterative guidance to progressively preserve background structures.

Due to the limitation of the receptive field size, CNN struggles to capture long-range dependencies in images, which can easily lead to poor overall visual effects in rain removal results. However, models based on the vision Transformer (ViT) architecture have overcome this limitation through their self-attention mechanisms. Xiao et al. [20] proposed an image deraining Transformer, which combines complementary window-based and spatial Transformers to capture local and long-range dependencies, while introducing a separate positional representation space and relative position enhanced multi-head self-attention to model positional relationships. Chen et al. [21] introduced a learnable top-k selection mechanism into self-attention, which adaptively retains only the most informative attention values for each query to suppress interference caused by irrelevant features and achieve more effective feature aggregation. Chen et al. [22] introduced a multi-scale Transformer framework which progressively reconstructs rain-free images from coarse to fine scales using multi-input and multi-output structure. Chen et al. [23] introduced implicit neural representations based on pixel coordinates to learn common degradation features of spatially varying rain streaks, and employed bidirectional cross-scale feedback to facilitate information exchange between coarse and fine scales. Zou et al. [24] proposed a cross-paradigm Transformer framework that jointly exploits spatial-channel and global-local representations, with complementary self-attention mechanisms and progressive feature alignment introduced to facilitate their interaction and fusion for image deraining. Li et al. [25] introduced a region-aware Transformer that separately models rain-affected and unaffected regions, employing region masked attention to selectively capture their individual features and cross-region interactions for more effective deraining.

Compared to ViT, Vision Mamba (ViM) based on the 2D State Space Model (2D-SSM) also exhibits excellent global modeling capabilities and lower computational overhead. Therefore, recent research efforts have begun to utilize ViM to construct image deraining networks. Shi et al. [26] proposed an omni selective scan mechanism that models image information along the horizontal, vertical, and channel dimensions, enabling comprehensive long-range feature modeling. Zou et al. [27] incorporated frequency analysis into Mamba to better model global rain degradation, using frequency-band decomposition for frequency correlation modeling and Fourier transformation for global feature representation. Gu et al. [28] replaced recursive state-space computation in Mamba with linear attention, exploiting their mathematical equivalence to retain global modeling while enabling more efficient computation. Jiang et al. [29] introduced position-guided unidirectional scanning into Mamba, which jointly encodes degradation features and spatial coordinates to selectively focus on degradation-relevant tokens, while harmonized attention further enhances spatial dependency modeling.

In addition to the above three common architectures, there are also some single-task image deraining models based on special architectures, such as spiking neural network [30], graph neural network [31], diffusion model [32] and deep unfolding network [33].

The aforementioned single-task image deraining models primarily focus on daytime scenes, while certain research efforts have specifically optimized for the characteristics of nighttime scenes. Zhang et al. [34] introduced a rain location prior learned implicitly from rainy images to characterize the spatial distribution of nighttime rain streaks. Guan et al. [35] proposed a learnable color space transformation to adaptively construct a more suitable luminance representation for nighttime image deraining. Liu et al. [36] designed a dual-prior-guided framework that disentangles lighting and rain degradation through Retinex-based decomposition.

Single-task image deraining models are well suited to scenarios with fixed and known rain types; however, in many practical situations, the rain type encountered by the model is unknown in advance and multiple rain types may occur sequentially, making it difficult to achieve timely and accurate model switching even when separate model weights are trained for different rain types.

C. Unified Image Deraining Models

Recent research efforts have begun to explore unified image deraining models, with the objective of training a singular model weight to simultaneously address various types of rain. Yan et al. [37] proposed the URIR-8K dataset for unified image deraining and developed a multi-scale Mamba that adopts scale-specific 2D scanning to better model rain patterns across different spatial scales. Jiao et al. [38] employed dual implicit neural representations in RGB and HSV color spaces to learn complementary features across diverse types of rain. Yang et al. [39] introduced the RainRAG dataset and developed an asymmetric mixture-of-experts architecture that employs multi-objective reweighted optimization to adaptively balance different rain types for unified deraining.

In fact, single-task image deraining models can also achieve a certain degree of unified deraining capability when retrained on newly proposed unified deraining datasets. However, their core designs are generally tailored to a specific rain type, making it difficult to achieve satisfactory unified deraining performance even after retraining. Although existing unified deraining methods have explored ways to accommodate various types of rain, they still suffer from limited degradation modeling capacity, gradient conflicts during training, and insufficient semantic discrimination, leaving considerable room for improvement.

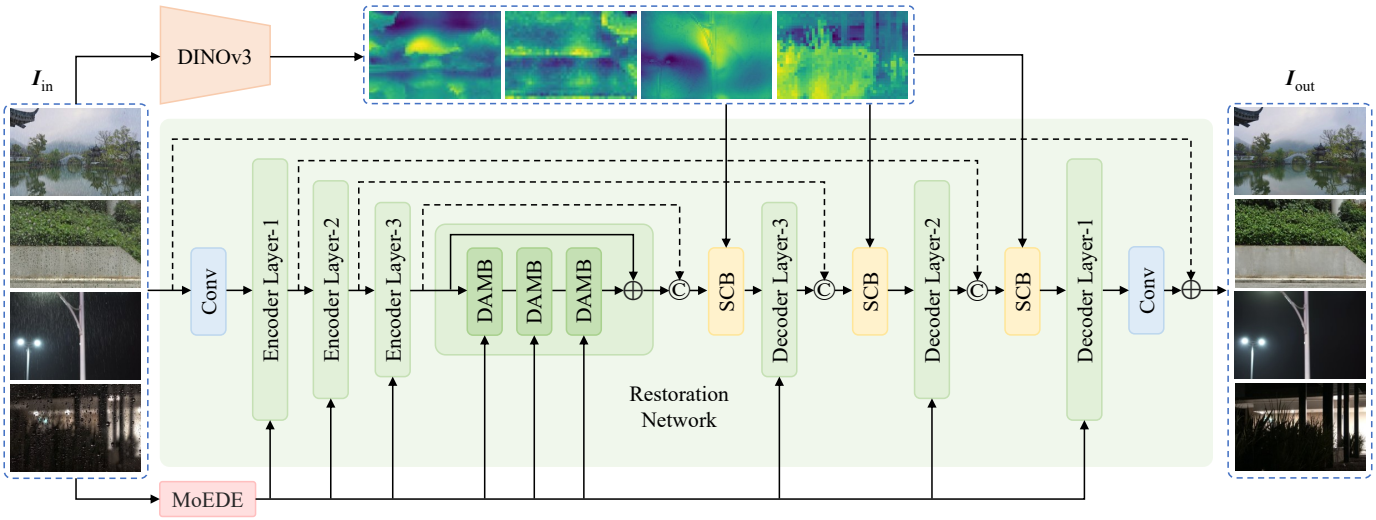


Fig. 2. Overall architecture of the proposed unified image deraining framework.

III. Methodology

A. Overview

Fig. 2 illustrates the overall architecture of the proposed unified image deraining framework, which consists of a mixture-of-experts degradation evaluator (MoEDE), a restoration network (RN) and a retrained DINOv3. The RN adopts a U-Net architecture, whose encoder and decoder are composed of ViM-based degradation-aware Mamba blocks (DAMBs). Downsampling and upsampling operations are performed between the DAMB groups to adjust the spatial resolutions and channel dimensions of feature maps. Given a rainy image, the MoEDE estimates two types of degradation representations from multiple perspectives and feeds them into the DAMBs to guide the restoration process. In parallel, the retrained DINOv3 recovers clean semantic representations directly from the rainy image and incorporates them into the decoder through semantic compensation blocks (SCBs), which modulate decoder features to improve rain-background discrimination and detail reconstruction. By integrating degradation guidance with semantic restoration, the RN progressively transforms the rainy input into a clean output.

B. Module Details

1) DAMB: As shown in Fig. 3(a), DAMB adopts a residual architecture consisting of a 2D-SSM and a feed-forward network (FFN). Given an input feature tensor $\boldsymbol{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively, layer normalization (LN) is first applied before the features are processed by the 2D-SSM. The resulting features are added to the original input through a residual connection to obtain the intermediate feature $\boldsymbol{F}_{\text{mid}}$:

$$\boldsymbol{F}_{\text{mid}} = \text{2D-SSM}\left(\text{LN}\left(\boldsymbol{F}_{\text{in}}\right)\right) + \boldsymbol{F}_{\text{in}}. \tag{1}$$

Subsequently, $\boldsymbol{F}_{\text{mid}}$ is normalized and passed through the FFN. Another residual connection is employed to obtain the output feature $\boldsymbol{F}_{\text{out}}$:

$$\boldsymbol{F}_{\text{out}} = \text{FFN}\left(\text{LN}\left(\boldsymbol{F}_{\text{mid}}\right)\right) + \boldsymbol{F}_{\text{mid}}. \tag{2}$$

The internal computation of the 2D-SSM is detailed in the dashed box. $\boldsymbol{F}_{\text{in}}$ is first normalized by LN and projected through a linear layer, after which the projected feature is equally split along the channel dimension into two branches:

$$[\boldsymbol{F}_1, \boldsymbol{F}_2] = \text{Split}\left(\text{Linear}\left(\text{LN}\left(\boldsymbol{F}_{\text{in}}\right)\right)\right). \tag{3}$$

In the first branch, depth-wise convolution (DConv) is followed by spatial scanning and an S6 block. LN is then applied to obtain the transformed feature $\boldsymbol{F}'_1$:

$$\boldsymbol{F}'_1 = \text{LN}\left(\text{S6}\left(\text{Scan}\left(\text{DConv}\left(\boldsymbol{F}_1\right)\right)\right)\right). \tag{4}$$

In the second branch, a linear transformation followed by SiLU activation produces the feature $\boldsymbol{F}'_2$:

$$\boldsymbol{F}'_2 = \text{SiLU}\left(\text{Linear}\left(\boldsymbol{F}_2\right)\right). \tag{5}$$

Finally, the two branch features are combined by element-wise multiplication and projected through a linear layer to obtain the 2D-SSM output $\boldsymbol{F}_{\text{SSM}}$:

$$\boldsymbol{F}_{\text{SSM}} = \text{Linear}\left(\boldsymbol{F}'_1 \odot \boldsymbol{F}'_2\right), \tag{6}$$

where $\text{Split}(\cdot)$ denotes channel-wise feature splitting, and $\odot$ denotes element-wise multiplication.

2) MoEDE: As illustrated in Fig. 3(b), MoEDE estimates the degradation representations of the input image $\boldsymbol{I}_{\text{in}} \in \mathbb{R}^{3\times H\times W}$ from multiple perspectives. A convolutional layer first extracts the shallow feature $\boldsymbol{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ from $\boldsymbol{I}_{\text{in}}$. $\boldsymbol{F}_{\text{in}}$ is then fed into a gating network consisting of global average pooling (GAP), a multilayer perceptron (MLP), and a Softmax function to generate three expert weights:

$$w_1, w_2, w_3 = \text{Softmax}\left(\text{MLP}\left(\text{GAP}\left(\boldsymbol{F}_{\text{in}}\right)\right)\right). \tag{7}$$

MoEDE contains three complementary experts: a local expert (LE) captures local spatial patterns, a global expert (GE) models long-range spatial dependencies, and a frequency expert (FE) extracts rain-related characteristics in the frequency domain. Their outputs are weighted by the corresponding

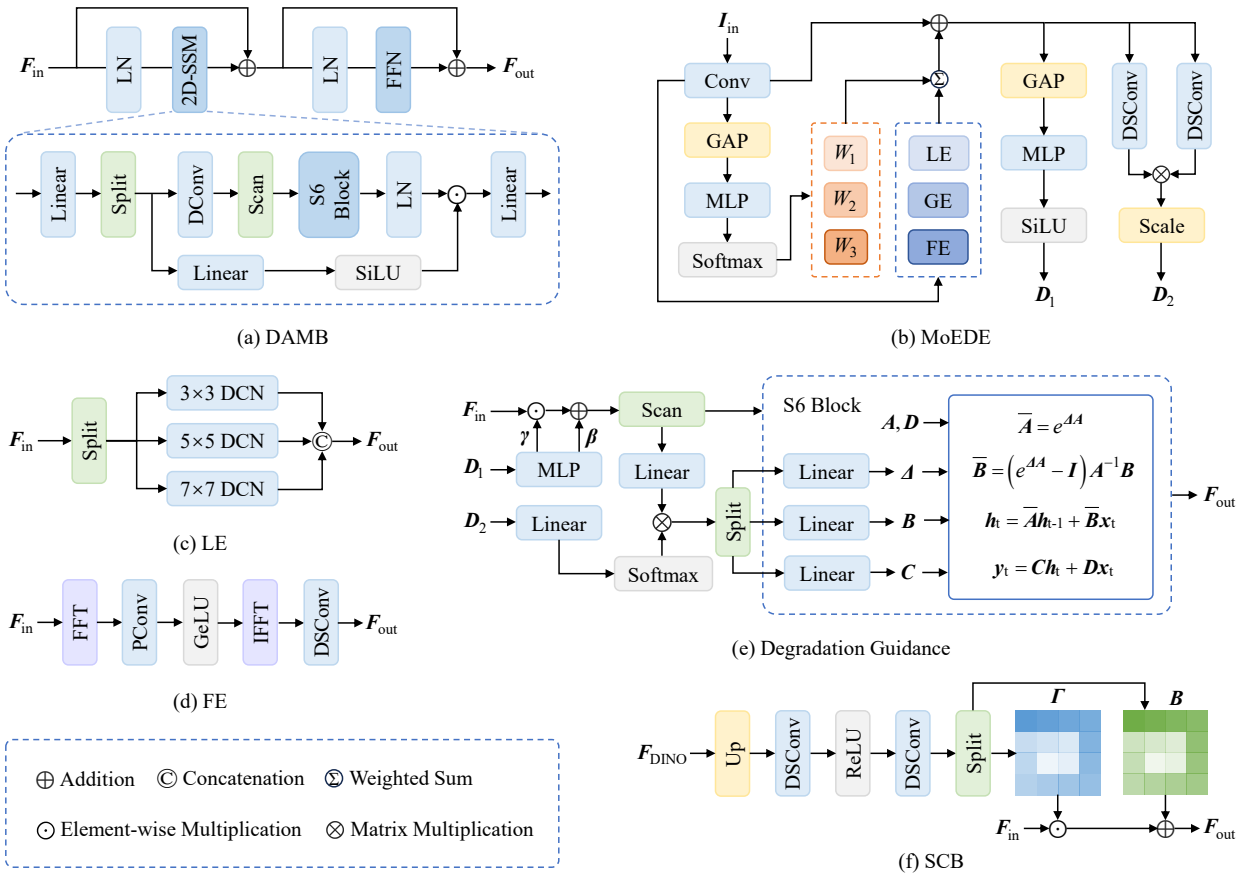


Fig. 3. Module details.

gating weights and fused with the original features through a residual connection:

$$\boldsymbol{F}_e = \boldsymbol{F}_{\text{in}} + \sum_{i=1}^{3} w_i \mathcal{E}_i\left(\boldsymbol{F}_{\text{in}}\right), \tag{8}$$

where $\mathcal{E}_i(\cdot)$ denotes the output feature of the i-th expert, and $\boldsymbol{F}_e$ is the fused expert feature.

LE divides $\boldsymbol{F}_{\text{in}}$ evenly into three groups along the channel dimension. These groups are processed by deformable convolutional networks (DCNs) with different kernel sizes to capture various local rain patterns, and the results are concatenated:

$$\begin{cases} \boldsymbol{F}_{\text{LE},1} = \text{DCN}_{3\times3}\left(\text{Split}(\boldsymbol{F}_{\text{in}})[1]\right), \\ \boldsymbol{F}_{\text{LE},2} = \text{DCN}_{5\times5}\left(\text{Split}(\boldsymbol{F}_{\text{in}})[2]\right), \\ \boldsymbol{F}_{\text{LE},3} = \text{DCN}_{7\times7}\left(\text{Split}(\boldsymbol{F}_{\text{in}})[3]\right), \\ \boldsymbol{F}_{\text{LE}} = \text{Concat}\left(\boldsymbol{F}_{\text{LE},1}, \boldsymbol{F}_{\text{LE},2}, \boldsymbol{F}_{\text{LE},3}\right). \end{cases} \tag{9}$$

Here, $\text{Split}(\cdot)[i]$ denotes the i-th group and $\text{Concat}(\cdot)$ denotes feature concatenation along the channel dimension.

GE shares a similar structure to DAMB, utilizing the long-range dependency modeling capability of ViM to extract global spatial features. Unlike DAMB, GE serves solely as a feature extractor without degradation guidance.

FE operates in the frequency domain. Specifically, the input features are transformed using the fast Fourier transform (FFT), followed by pixel-wise convolution (PConv) and GeLU activation. The resulting frequency features are then transformed back to the spatial domain using the inverse fast Fourier transform (IFFT) and processed by a depth-wise separable convolution (DSConv):

$$\begin{cases} \boldsymbol{F}_f = \text{GeLU}\left(\text{PConv}\left(\text{FFT}(\boldsymbol{F}_{\text{in}})\right)\right), \\ \boldsymbol{F}_{\text{FE}} = \text{DSConv}\left(\text{IFFT}(\boldsymbol{F}_f)\right). \end{cases} \tag{10}$$

Based on $\boldsymbol{F}_e$, MoEDE generates two types of degradation representation. First, GAP, MLP and SiLU activation are applied to obtain 1D degradation representation $\boldsymbol{D}_1 \in \mathbb{R}^L$:

$$\boldsymbol{D}_1 = \text{SiLU}\left(\text{MLP}\left(\text{GAP}\left(\boldsymbol{F}_e\right)\right)\right). \tag{11}$$

This representation implicitly encodes the overall rain type. Then, two DSConv operations are applied to $\boldsymbol{F}_e$ to obtain two feature matrices, their matrix product is normalized by $H \times W$ to compensate for the variations in image sizes, yielding 2D degradation representation $\boldsymbol{D}_2 \in \mathbb{R}^{L\times L}$:

$$\boldsymbol{D}_2 = \frac{\text{DSConv}_1(\boldsymbol{F}_e) \otimes \text{DSConv}_2(\boldsymbol{F}_e)}{H \times W}. \tag{12}$$

This representation aims to characterize the spatial distribution of rain. Together, $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ provide complementary degradation information of the input rainy image to guide the RN subsequently.

3) Degradation Guidance: As illustrated in Fig. 3(e), $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are incorporated into the DAMB to adapt its feature transformation and state-space computation to the degradation characteristics of the input rainy image.

First, $\boldsymbol{D}_1$ is mapped through MLP to generate channel-wise scaling and bias vectors, which are used to perform an affine transformation on the input features:

$$\begin{cases} \boldsymbol{\gamma}, \boldsymbol{\beta} = \mathrm{MLP}(\boldsymbol{D}_1), \\ \boldsymbol{F}'_{\mathrm{in}} = \boldsymbol{F}_{\mathrm{in}} \odot \boldsymbol{\gamma} + \boldsymbol{\beta}, \end{cases} \tag{13}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^C$. The modulated feature $\boldsymbol{F}'_{\mathrm{in}}$ is then processed by pixel-wise scanning and a linear projection to obtain $\boldsymbol{F}_{\mathrm{scan}} \in \mathbb{R}^{C \times HW}$:

$$\boldsymbol{F}_{\mathrm{scan}} = \mathrm{Linear}\left(\mathrm{Scan}(\boldsymbol{F}'_{\mathrm{in}})\right). \tag{14}$$

Different channels of $\boldsymbol{F}_{\mathrm{scan}}$ can be regarded as focusing on different regions or specific structures of the image.

Next, $\boldsymbol{D}_2$ is mapped through a linear layer followed by Softmax to produce channel-wise scanning weights $\boldsymbol{W}_{\mathrm{scan}} \in \mathbb{R}^C$. These weights are assigned to the corresponding channels of $\boldsymbol{F}_{\mathrm{scan}}$ to obtain the degradation-conditioned feature $\boldsymbol{F}_m$:

$$\begin{cases} \boldsymbol{W}_{\mathrm{scan}} = \mathrm{Softmax}\left(\mathrm{Linear}(\boldsymbol{D}_2)\right), \\ \boldsymbol{F}_m = \boldsymbol{F}_{\mathrm{scan}} \odot \boldsymbol{W}_{\mathrm{scan}}. \end{cases} \tag{15}$$

To dynamically construct the key parameters of the S6 block, $\boldsymbol{F}_m$ is evenly divided into three groups along the channel dimension. Each group is independently mapped through a linear layer to generate the corresponding parameters:

$$\begin{cases} \boldsymbol{\Delta} = \mathrm{Linear}\left(\mathrm{Split}(\boldsymbol{F}_m)[1]\right), \\ \boldsymbol{B} = \mathrm{Linear}\left(\mathrm{Split}(\boldsymbol{F}_m)[2]\right), \\ \boldsymbol{C} = \mathrm{Linear}\left(\mathrm{Split}(\boldsymbol{F}_m)[3]\right). \end{cases} \tag{16}$$

The state-space parameters $\boldsymbol{A}$ and $\boldsymbol{D}$ retain the default initialization used in ViM.

Through these two complementary guidance pathways, $\boldsymbol{D}_1$ modulates channel-wise feature transformations, while $\boldsymbol{D}_2$ dynamically determines the key parameters of the S6 block. This design enables the restoration network to adapt its state-space computation to different rain degradations, thereby mitigating gradient conflicts that may arise when a single network is trained to handle heterogeneous rain types.

4) SCB: As shown in Fig. 3(f), SCB exploits the clean semantic representation $\boldsymbol{F}_{\mathrm{DINO}}$ recovered by DINOv3 to modulate the decoder features in the restoration network. Given the intermediate decoder feature $\boldsymbol{F}_{\mathrm{in}} \in \mathbb{R}^{C \times H \times W}$, $\boldsymbol{F}_{\mathrm{DINO}}$ is first upsampled to match the spatial resolution of $\boldsymbol{F}_{\mathrm{in}}$, and then processed by two DSConvs with ReLU activation in between, producing $\boldsymbol{F}'_{\mathrm{DINO}} \in \mathbb{R}^{2C \times H \times W}$:

$$\boldsymbol{F}'_{\mathrm{DINO}} = \mathrm{DSConv}\left(\mathrm{ReLU}\left(\mathrm{DSConv}\left(\mathrm{Up}(\boldsymbol{F}_{\mathrm{DINO}})\right)\right)\right). \tag{17}$$

Next, $\boldsymbol{F}'_{\mathrm{DINO}}$ is evenly split along the channel dimension into a scaling tensor $\boldsymbol{\Gamma}$ and a bias tensor $\boldsymbol{B}$:

$$\boldsymbol{\Gamma}, \boldsymbol{B} = \mathrm{Split}(\boldsymbol{F}'_{\mathrm{DINO}}), \tag{18}$$

where $\boldsymbol{\Gamma}, \boldsymbol{B} \in \mathbb{R}^{C \times H \times W}$. They are then used to perform a pixel-wise affine transformation on the decoder feature, resulting in the output feature $\boldsymbol{F}_{\mathrm{out}} \in \mathbb{R}^{C \times H \times W}$:

$$\boldsymbol{F}_{\mathrm{out}} = \boldsymbol{F}_{\mathrm{in}} \odot \boldsymbol{\Gamma} + \boldsymbol{B}. \tag{19}$$

By injecting clean semantic representations into the decoder features, SCB provides complementary semantic cues for distinguishing rain from background structures, thereby improving rain removal and background detail reconstruction.

C. Training Strategy

In other research avenues within the realm of image enhancement and restoration, certain methods [43]–[45] have shifted away from using clean images exclusively as supervisory signals. Instead, they leverage the discrepancies between clean images and degraded images to extract specific prior information, thereby guiding the restoration network. Inspired by these methods, a three-stage teacher-student training strategy is proposed as shown in Fig. 4.

1) Stage 1: The first training stage jointly optimizes a teacher MoEDE and the RN to learn degradation representations that facilitate image deraining. Meanwhile, a frozen pretrained DINOv3 provides clean semantic representations for the restoration process. Given the input rainy image $\boldsymbol{I}_{\mathrm{in}}$ and its ground-truth $\boldsymbol{I}_{\mathrm{gt}}$, the teacher MoEDE takes both images as input and generates 1D and 2D degradation representations:

$$\boldsymbol{D}_1, \boldsymbol{D}_2 = \text{MoEDE-T}\left(\boldsymbol{I}_{\mathrm{in}}, \boldsymbol{I}_{\mathrm{gt}}\right), \tag{20}$$

Meanwhile, a pretrained DINOv3 is kept frozen and used as the teacher DINOv3. Taking the clean image $\boldsymbol{I}_{\mathrm{gt}}$ as input, it extracts clean semantic representations:

$$\boldsymbol{F}_{\mathrm{DINO}} = \text{DINOv3-T}\left(\boldsymbol{I}_{\mathrm{gt}}\right). \tag{21}$$

The RN takes $\boldsymbol{I}_{\mathrm{in}}$, $\boldsymbol{D}_1$, $\boldsymbol{D}_2$, and $\boldsymbol{F}_{DINO}$ as input to produce a clean-image-assisted deraining result:

$$\boldsymbol{I}_{\mathrm{out1}} = \mathrm{DN}\left(\boldsymbol{I}_{\mathrm{in}}, \boldsymbol{D}_1, \boldsymbol{D}_2, \boldsymbol{F}_{\mathrm{DINO}}\right). \tag{22}$$

To supervise image reconstruction, a pixel-wise loss is imposed between $\boldsymbol{I}_{\mathrm{out1}}$ and $\boldsymbol{I}_{\mathrm{gt}}$:

$$\mathcal{L}_{\mathrm{pixel}}(\boldsymbol{I}_{\mathrm{out1}}, \boldsymbol{I}_{\mathrm{gt}}) = \left\|\boldsymbol{I}_{\mathrm{out1}} - \boldsymbol{I}_{\mathrm{gt}}\right\|_1. \tag{23}$$

In addition, a Pearson correlation loss $\mathcal{L}_{\mathrm{cor}}$ [46] is introduced to regulate the linear relationship between $\boldsymbol{I}_{\mathrm{out1}}$ and $\boldsymbol{I}_{\mathrm{gt}}$. The overall objective of the first training stage is defined as:

$$\mathcal{L}_{\mathrm{stage1}} = \mathcal{L}_{\mathrm{pixel}}(\boldsymbol{I}_{\mathrm{out1}}, \boldsymbol{I}_{\mathrm{gt}}) + \mathcal{L}_{\mathrm{cor}}(\boldsymbol{I}_{\mathrm{out1}}, \boldsymbol{I}_{\mathrm{gt}}). \tag{24}$$

By jointly optimizing the teacher MoEDE and the RN under this constraint, the first training stage establishes reference degradation representations and equips the RN with clean semantic representation.

2) Stage 2: Semantic representations extracted from rainy images by the original pretrained DINOv3 contain degradation-related interference, which may impair semantic compensation in the RN. However, only rainy images are available during inference. Therefore, the second training stage retrains a student DINOv3 based on the pretrained weights to recover clean semantic representations from rainy images.

Given a rainy image $\boldsymbol{I}_{\mathrm{in}}$ and its ground-truth $\boldsymbol{I}_{\mathrm{gt}}$, the frozen teacher DINOv3 extracts the reference representation from $\boldsymbol{I}_{\mathrm{gt}}$, while the student DINOv3 predicts the corresponding representation from $\boldsymbol{I}_{\mathrm{in}}$:

$$\begin{cases} \boldsymbol{F}_{\mathrm{DINO}} = \text{DINOv3-T}\left(\boldsymbol{I}_{\mathrm{gt}}\right), \\ \hat{\boldsymbol{F}}_{\mathrm{DINO}} = \text{DINOv3-S}\left(\boldsymbol{I}_{\mathrm{in}}\right). \end{cases} \tag{25}$$

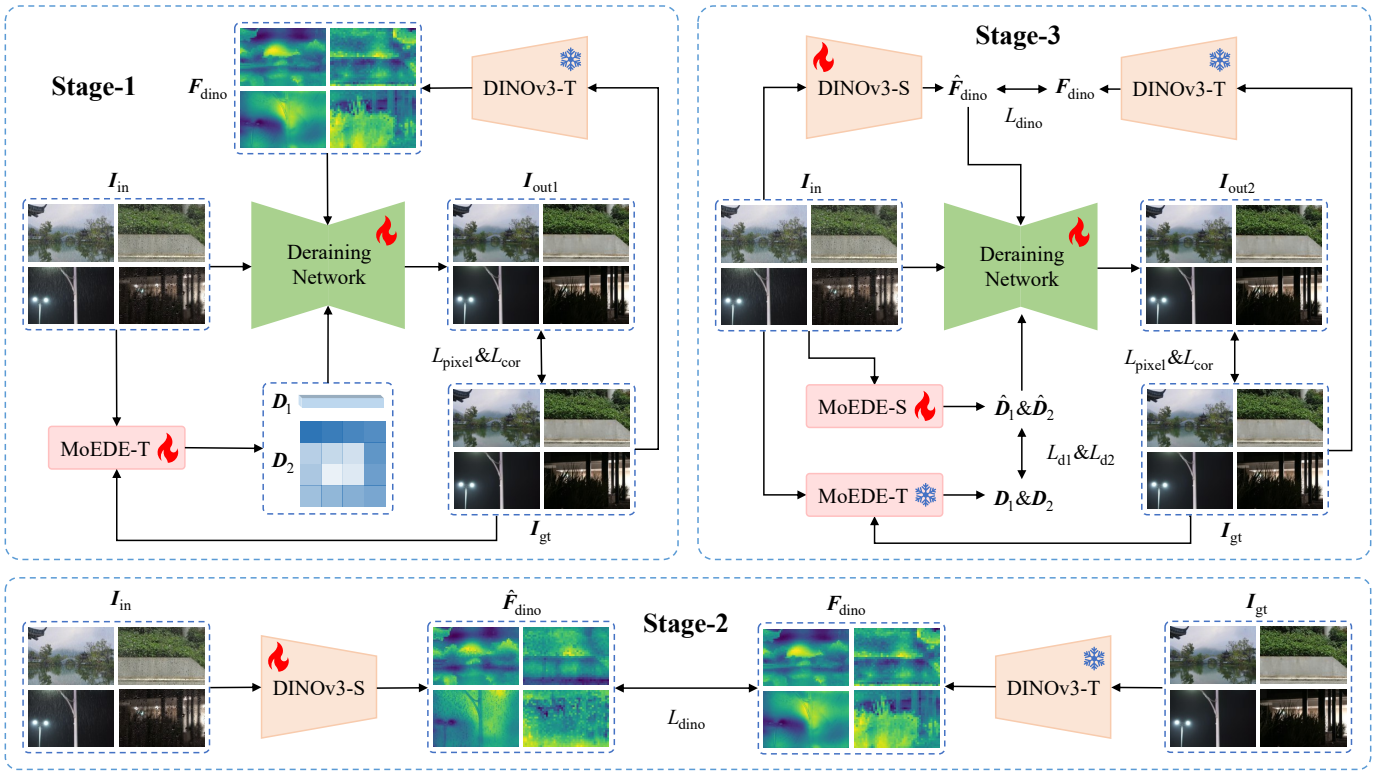


Fig. 4. Training strategy.

The student DINOv3 is optimized to match the reference representation through ℓ_1 loss:

$$\mathcal{L}_{\mathrm{DINO}} = \left\| \hat{\boldsymbol{F}}_{\mathrm{DINO}} - \boldsymbol{F}_{\mathrm{DINO}} \right\|_1 . \tag{26}$$

By learning to reproduce clean semantic representations from rainy images, the student DINOv3 acquires the ability to provide clean semantic cues without clean images at inference.

3) Stage 3: The third training stage introduces a student MoEDE to estimate degradation representations directly from rainy images. Given a rainy image $\boldsymbol{I}_{\mathrm{in}}$, the student MoEDE predicts the 1D degradation representation $\hat{\boldsymbol{D}}_1$ and 2D degradation representation $\hat{\boldsymbol{D}}_2$:

$$\hat{\boldsymbol{D}}_1, \hat{\boldsymbol{D}}_2 = \text{MoEDE-S}\left(\boldsymbol{I}_{\mathrm{in}}\right). \tag{27}$$

The degradation representations $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ obtained in the first stage serve as reference targets for training the student MoEDE. Specifically, a cosine similarity loss is imposed on the 1D degradation representation, while an element-wise ℓ_1 loss is used for the 2D degradation representations:

$$\begin{cases} \mathcal{L}_{\mathrm{d1}} = 1 - \dfrac{\hat{\boldsymbol{D}}_1^{\top} \boldsymbol{D}_1}{\|\hat{\boldsymbol{D}}_1\|_2 \|\boldsymbol{D}_1\|_2}, \\ \mathcal{L}_{\mathrm{d2}} = \left\| \hat{\boldsymbol{D}}_2 - \boldsymbol{D}_2 \right\|_1 . \end{cases} \tag{28}$$

The RN is initialized with the weights learned in the first stage and jointly optimized with the student MoEDE and the student DINOv3. Given the rainy image $\boldsymbol{I}_{\mathrm{in}}$, the newly predicted degradation representations $\hat{\boldsymbol{D}}_1$&$\hat{\boldsymbol{D}}_2$, and the recovered clean semantic representation $\hat{\boldsymbol{F}}_{\mathrm{DINO}}$, the RN produces the final derained image $\boldsymbol{I}_{\mathrm{out2}}$:

$$\boldsymbol{I}_{\mathrm{out2}} = \mathrm{DN}\left(\boldsymbol{I}_{\mathrm{in}}, \hat{\boldsymbol{D}}_1, \hat{\boldsymbol{D}}_2, \hat{\boldsymbol{F}}_{\mathrm{DINO}}\right). \tag{29}$$

The overall objective of the third stage combines the image reconstruction loss and the representation reproduction loss:

$$\begin{aligned} \mathcal{L}_{\mathrm{stage3}} = \mathcal{L}_{\mathrm{pixel}}\left(\boldsymbol{I}_{\mathrm{out2}}, \boldsymbol{I}_{\mathrm{gt}}\right) + \mathcal{L}_{\mathrm{cor}}\left(\boldsymbol{I}_{\mathrm{out2}}, \boldsymbol{I}_{\mathrm{gt}}\right) \\ + \mathcal{L}_{\mathrm{d1}} + \mathcal{L}_{\mathrm{d2}} + \mathcal{L}_{\mathrm{DINO}}. \end{aligned} \tag{30}$$

Through this joint optimization, the final unified deraining model can perform degradation-guided restoration and semantic compensation using rainy images alone during inference, without requiring clean images.

IV. Experiments

A. Experiment Setting

1) Datasets: RainRAG dataset [39] is used as the sole training dataset for the proposed method and all competing methods. It contains paired rainy and clean images covering four rain types: daytime rain streak (DRS), daytime raindrop (DRD), nighttime rain streak (NRS), and nighttime raindrop (NRD). The training set consists of 52,869 image pairs, while the test set is divided into four subsets corresponding to the respective rain types, with 100 image pairs in each subset. To evaluate cross-dataset generalization performance, all trained models are further tested on four real-world rainy image datasets, namely RealRain-1k-L [47], RealRain-1k-H [47], WeatherBench-rain [48] and RainDS-real-RD [49]. In addition, real-world rainy photographs collected from the

TABLE I
QUANTITATIVE COMPARISONS OF DERAINING METHODS ON FOUR RAINRAG TEST SUBSETS [39]. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. ↑ INDICATES THAT HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE, WHILE ↓ INDICATES THAT LOWER VALUES CORRESPOND TO BETTER PERFORMANCE

Methods	Venue	DRS			DRD			NRS			NRD		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
IDT [20]	TPAMI 2022	26.97	0.8279	0.1960	23.17	0.7648	0.2300	33.05	0.9470	0.2132	25.20	0.7963	0.3718
RLP [34]	ICCV 2023	28.01	0.8290	0.2339	22.42	0.7353	0.3029	32.06	0.9438	0.1971	23.48	0.7638	0.4282
DRSformer [21]	CVPR 2023	28.08	0.8253	0.1960	22.93	0.7499	0.2407	32.86	0.9457	0.2202	24.99	0.7953	0.3702
MSDT [22]	AAAI 2024	28.60	0.8414	0.1576	23.31	0.7487	0.2587	34.56	0.9591	0.1677	25.28	0.8148	0.3555
NeRD-Rain [23]	CVPR 2024	28.11	0.8309	0.1720	23.30	0.7513	0.2318	33.88	0.9511	0.1964	25.31	0.8027	0.3571
URIR [37]	AAAI 2025	28.29	0.8386	0.2001	23.19	0.7483	0.2815	34.32	0.9570	0.1860	25.82	0.8263	0.3390
CPRAformer [24]	ACM MM 2025	30.11	0.8464	**0.1376**	24.84	0.7791	0.2020	35.69	0.9635	0.1477	26.29	0.8342	0.3280
NDMamba [36]	TIP 2025	30.03	0.8439	0.1432	25.01	0.7795	0.2077	36.40	0.9649	0.1447	26.46	0.8375	0.3267
Regformer [25]	TMM 2026	29.98	0.8510	0.1487	24.98	0.7825	0.2100	35.96	0.9654	0.1460	26.26	0.8359	0.3328
UniRain [39]	CVPR 2026	29.58	0.8400	0.1814	24.71	0.7687	0.2408	35.23	0.9592	0.1617	26.21	0.8380	0.3275
Ours	–	**30.28**	**0.8516**	0.1455	**25.42**	**0.7898**	**0.1916**	**37.20**	**0.9687**	**0.1345**	**28.06**	**0.8628**	**0.3046**

Internet, for which clean images are unavailable, are used for further qualitative evaluation of generalization performance.

2) Implementation Details: The proposed method was trained on a single NVIDIA GeForce RTX 4090D GPU. The AdamW optimizer was employed with an initial learning rate of 3×10^{-4}, which was gradually reduced to 1×10^{-5} using a cosine annealing schedule over approximately 370,000 iterations. Training patches were randomly cropped from the training images and a progressive training strategy [50] was adopted: batch size and patch resolution were successively set to 6 with 160×160, 4 with 192×192, 2 with 272×272, and 1 with 384×384. The dimension L of the degradation representations was set to 96, and the ViT-S variant of DINOv3 was used to extract the semantic representation.

3) Comparative Methods and Evaluation Metrics: The proposed method was compared with ten SOTA methods, including IDT [20], RLP [34], DRSformer [21], MSDT [22], NeRD-Rain [23], URIR [37], CPRAformer [24], NDMamba [36], Regformer [25] and UniRain [39]. We used the Peak Signal to Noise Ratio (PSNR), Structure Similarity Index Measure (SSIM) [51] and Learned Perceptual Image Patch Similarity (LPIPS) [52] to evaluate the deraining performance.

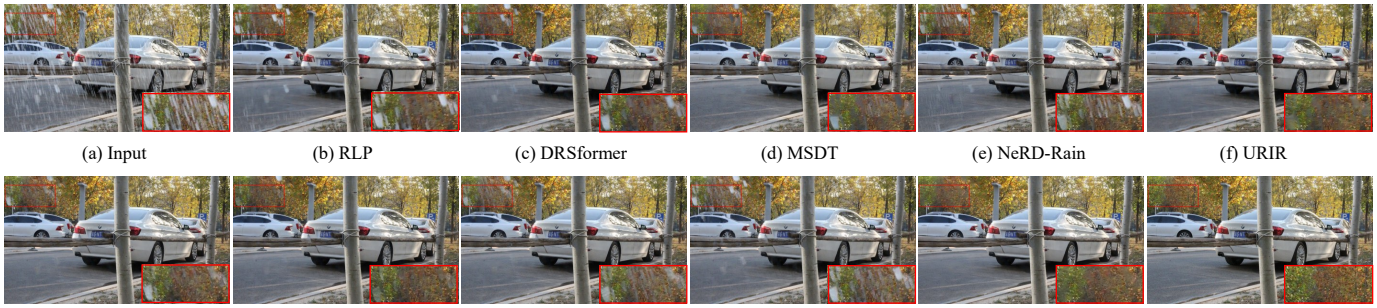
B. Comparison With SOTA Methods

1) Experiment Results on RainRAG Dataset: As shown in Table I, except for the LPIPS metric on the DRS test subset, the proposed model outperforms others on the remaining 11 metrics. Compared to the two daytime test subsets DRS and DRD, the proposed model demonstrates a more pronounced performance imporovement on the two nighttime test subsets NRS and NRD. As shown in Table II, across the four distinct test subsets, the proposed model achieves an average PSNR metric of 30.24dB, marking a 0.76dB improvement over the second-best value, equivalent to a 2.58% increase; it attains an average SSIM metric of 0.8682, showing a 0.0095 improvement over the second-best value, or a 1.11% increase; and records an average LPIPS metric of 0.1941, representing a 0.0097 decrease from the second-best value, yet indicating a 4.76% improvement in performance. In terms of model complexity, the proposed model contains a retrained DINOv3 (ViT-S version with 21.6M parameters and 5.6G FLOPs), so the total number of parameters is relatively large, reaching 35.8M. However, the total computational cost of 76.8G FLOPs is relatively low, ranking fourth among the 11 models.

TABLE II
QUANTITATIVE COMPARISONS OF THE AVERAGE PERFORMANCE ON FOUR RAINRAG TEST SUBSETS [39] AND MODEL COMPLEXITY. FLOPS ARE MEASURED AT A RESOLUTION OF 256×256

Methods	PSNR↑	SSIM↑	LPIPS↓	Params	FLOPs
IDT [20]	27.10	0.8340	0.2528	16.4M	61.9G
RLP [34]	26.49	0.8180	0.2905	5.5M	63.4G
DRSformer [21]	27.22	0.8291	0.2568	33.7M	242.9G
MSDT [22]	27.94	0.8410	0.2349	16.6M	129.9G
NeRD-Rain [23]	27.65	0.8340	0.2393	22.9M	156.3G
URIR [37]	27.91	0.8426	0.2517	11.8M	99.2G
CPRAformer [24]	29.23	0.8558	0.2038	29.3M	150.1G
NDMamba [36]	29.48	0.8565	0.2056	5.6M	27.4G
Regformer [25]	29.30	0.8587	0.2094	22.3M	184.4G
UniRain [39]	28.93	0.8515	0.2279	28.1M	204.0G
Ours	**30.24**	**0.8682**	**0.1941**	35.8M	76.8G

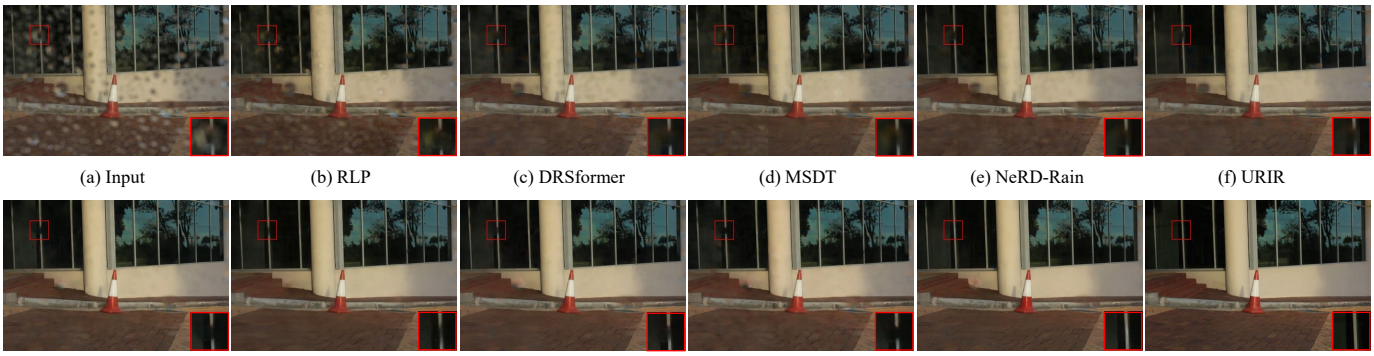
Qualitative comparisons of image deraining results are shown in Fig. 5 to Fig. 8. It can be observed that the proposed model has the following advantages. First, competing models often struggle to rain streaks with varying lengths and thicknesses, while the proposed model exhibits excellent removal capabilities for different rain forms. Second, after removing large raindrops, competing models often struggle to reconstruct the eroded background information, while the proposed model possesses strong contextual reasoning capabilities and excels in restoring texture details. Third, competing models often perform well only on specific rain types, while the proposed model demonstrates a more comprehensive and integrated image deraining performance.



(a) Input (b) RLP (c) DRSformer (d) MSDT (e) NeRD-Rain (f) URIR

(g) CPRAformer (h) NDMamba (i) Regformer (j) UniRain (k) Ours (l) GT

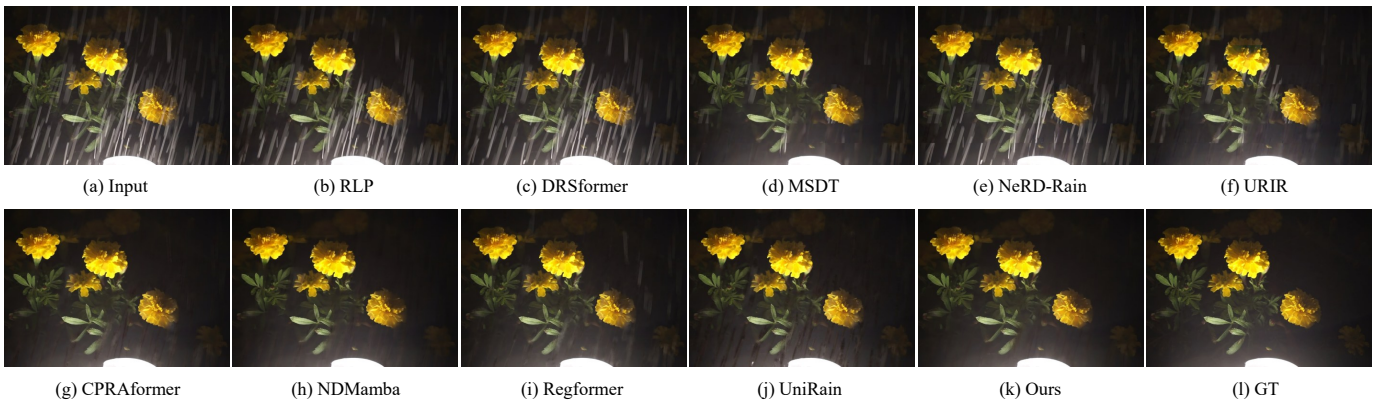
Fig. 5. Visual comparison of the deraining results on the DRS test subsets. Zoom in for a better view.



(a) Input (b) RLP (c) DRSformer (d) MSDT (e) NeRD-Rain (f) URIR

(g) CPRAformer (h) NDMamba (i) Regformer (j) UniRain (k) Ours (l) GT

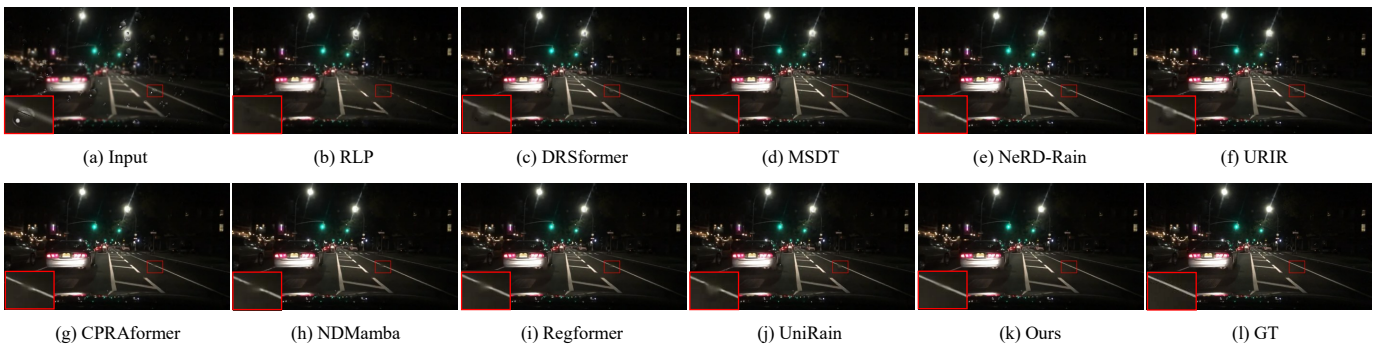
Fig. 6. Visual comparison of the deraining results on the DRD test subsets. Zoom in for a better view.



(a) Input (b) RLP (c) DRSformer (d) MSDT (e) NeRD-Rain (f) URIR

(g) CPRAformer (h) NDMamba (i) Regformer (j) UniRain (k) Ours (l) GT

Fig. 7. Visual comparison of the deraining results on the NRS test subsets. Zoom in for a better view.



(a) Input (b) RLP (c) DRSformer (d) MSDT (e) NeRD-Rain (f) URIR

(g) CPRAformer (h) NDMamba (i) Regformer (j) UniRain (k) Ours (l) GT

Fig. 8. Visual comparison of the deraining results on the NRD test subsets. Zoom in for a better view.

TABLE III
QUANTITATIVE COMPARISONS OF THE CROSS-DATASET GENERALIZATION TEST ON FOUR DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. ↑ INDICATES THAT HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE

Method	Venue	RealRain-1k-L		RealRain-1k-H		WeatherBench-rain		RainDS-real-RD		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
IDT [20]	TPAMI 2022	33.71	0.9664	30.37	0.9441	32.05	0.9355	20.35	0.6237	29.12	0.8674
RLP [34]	ICCV 2023	32.02	0.9558	29.01	0.9309	31.48	0.9330	20.86	0.6136	28.34	0.8583
DRSformer [21]	CVPR 2023	33.65	0.9625	30.35	0.9360	31.90	0.9281	20.15	0.6135	29.01	0.8600
MSDT [22]	AAAI 2024	33.56	0.9640	30.91	0.9451	33.56	0.9453	20.72	0.6250	29.69	0.8699
NeRD-Rain [23]	CVPR 2024	33.98	0.9710	31.37	0.9583	32.77	0.9372	21.01	0.6155	29.78	0.8705
URIR [37]	AAAI 2025	33.54	0.9686	31.10	0.9550	33.50	0.9435	20.44	0.5945	29.65	0.8654
CPRAformer [24]	ACM MM 2025	37.07	0.9804	33.98	0.9690	35.16	0.9519	21.64	0.6409	31.96	0.8856
NDMamba [36]	TIP 2025	37.55	0.9800	35.16	0.9727	35.58	0.9545	**22.37**	**0.6501**	32.67	0.8893
Regformer [25]	TMM 2026	36.97	0.9803	34.34	0.9711	35.64	0.9544	21.84	0.6388	32.20	0.8862
UniRain [39]	CVPR 2026	36.51	0.9778	33.74	0.9672	34.25	0.9448	22.07	0.6436	31.64	0.8834
Ours	–	**37.56**	**0.9834**	**36.02**	**0.9790**	**36.62**	**0.9593**	22.21	0.6399	**33.10**	**0.8904**



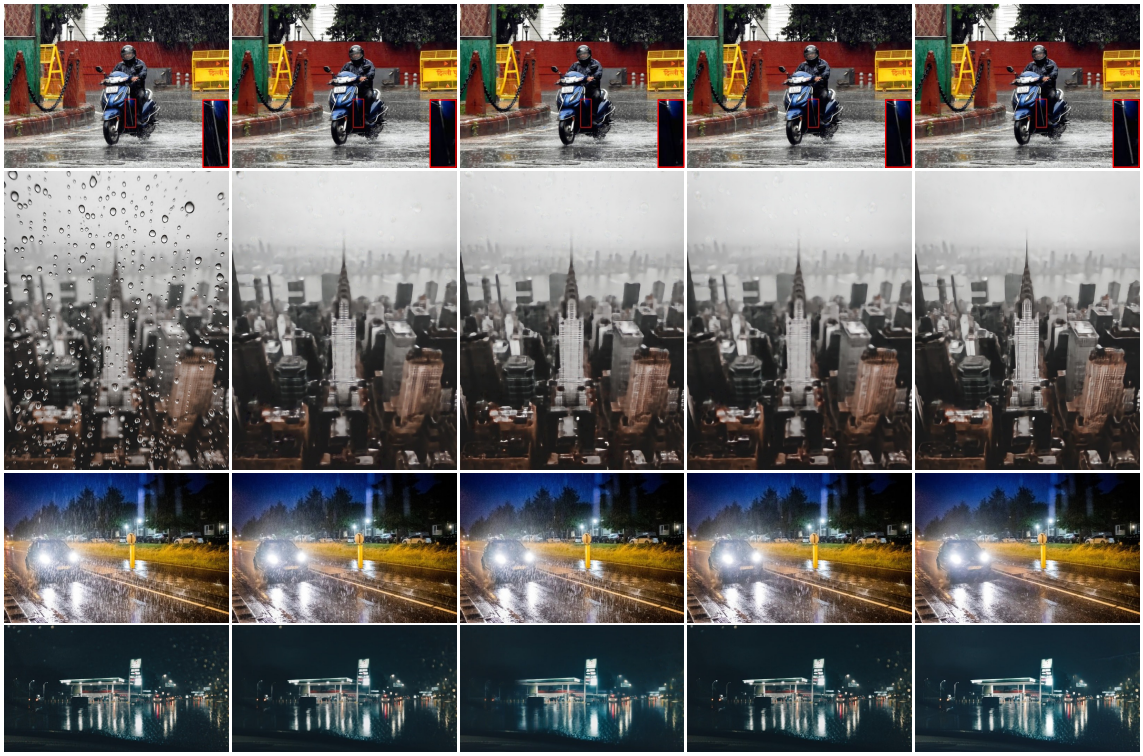
(a) Input (b) CPRAformer (c) NDMamba (d) Regformer (e) Ours (f) GT

Fig. 9. Visual comparison of the cross-dataset generalization test. The four rows of images from top to bottom are respectively selected from the RealRain-1k-L [47], RealRain-1k-H [47], WeatherBench-rain [48] and RainDS-real-RD [49]. Zoom in for a better view.

2) Cross-dataset Generalization Test: To evaluate the generalization ability of different methods, a cross-dataset test was conducted using models trained on the RainRAG dataset [39] and evaluated on four real-world deraining datasets: RealRain-1k-L [47], RealRain-1k-H [47], WeatherBench-rain [48] and RainDS-real-RD [49]. Quantitative comparisons are reported in Table III. It can be observed that the proposed method outperforms the competing methods on all metrics except for those on RainDS-real-RD [49], where its performance is slightly inferior to that of NDMamba [36].

Visual comparisons of cross-dataset deraining results are presented in Fig. 9. In the first row, the competing methods effectively remove rain streaks in the sky but leave residual rainwater flowing down from the eaves, while the proposed method handles this challenging case more effectively. In the second and third rows, the competing methods tend to leave streak-like artifacts or introduce color shifts after removing rain streaks around light sources, whereas such issues are rarely observed with the proposed method. In the fourth row, none of the methods can accurately restore background details obscured by dense and large raindrops, the proposed method produces slightly better overall visual quality.

3) Further Real-World Generalization Test: To further evaluate the generalization ability of the proposed model in real-world scenarios, rainy photos covering diverse scene types were collected from the Internet. Fig. 10 presents a visual comparison of the deraining results produced by the proposed model and three competitive methods. In the first row, all the



(a) Input (b) CPRAformer (c) NDMamba (d) Regformer (e) Ours

Fig. 10. Visual comparison of the further real-world generalization test. The input images are real-world rainy photos selected from the Internet while ground-truth clean images are unavailable. Zoom in for a better view.

methods effectively remove rain streaks. However, the vehicle edge within the red box looks like a rain streak, which causes CPRAformer [24] and NDMamba [36] to erroneously remove it, while Regformer [25] partially damages it. In contrast, the proposed model better distinguishes rain streaks from visually similar background structures by leveraging the semantic perception capability of DINOv3. In the second row, visible raindrop residues remain in the top of the images restored by all three competing methods. The proposed model avoids these artifacts and also partially compensates for background blur, producing greater sharpness and contrast. In the third row, the proposed model achieves superior performance in removing nighttime rain streaks compared to competing methods. In the fourth row, CPRAformer [24] and Regformer [25] exhibit limited effectiveness in removing nighttime raindrops. Although NDMamba [36] removes the raindrops relatively well, it also causes substantial damage to the background details, particularly the streetlight and the buildings on the left side of the image. In comparison, only the proposed model effectively balances raindrop removal and background preservation.

C. Ablation Studies

1) Module Design: To evaluate the effectiveness of the proposed modules, the following ablation experiments were conducted on the RainRAG dataset [39]. First, MoEDE, DINOv3, and SCB were removed to establish a RN without degradation guidance or semantic compensation. Starting from this baseline, the newly introduced modules were progressively incorporated to assess their respective contributions.

TABLE IV
ABLATION STUDY ON THE PROPOSED MODULES, ONLY THE AVERAGE METRICS ACROSS THE FOUR TEST SUBSETS [39] ARE REPORTED

Models	$\boldsymbol{D}_1$	$\boldsymbol{D}_2$	MoEDE	DINOv3	PSNR↑	SSIM↑	LPIPS↓
Baseline	✗	✗	✗	✗	29.60	0.8595	0.2045
$\mathcal{M}_1$	✓	✗	✗	✗	29.76	0.8616	0.2030
$\mathcal{M}_2$	✓	✓	✗	✗	29.97	0.8629	0.1998
$\mathcal{M}_3$	✓	✓	✓	✗	<u>30.10</u>	<u>0.8660</u>	<u>0.1974</u>
Ours	✓	✓	✓	✓	**30.24**	**0.8682**	**0.1941**

Specifically, $\mathcal{M}_1$ adds a degradation evaluator without expert networks to provide RN with 1D degradation representation. $\mathcal{M}_2$ further introduces 2D degradation representation based on $\mathcal{M}_1$. $\mathcal{M}_3$ incorporates expert networks into the degradation evaluator to form MoEDE. The proposed model is obtained by further adding DINOv3 and SCB to $\mathcal{M}_3$. As shown in Table IV, progressively incorporating the proposed modules into the baseline consistently improves the evaluation metrics.

2) Training Strategy of Degradation Guidance: To verify the validity of the proposed degradation guidance mechanism in training strategy, the following ablations were conducted:

- **Strategy 1.** No constraints are imposed on either type of degradation representation during training stage 3.
- **Strategy 2.** The loss constraint is imposed only on the 1D degradation representation during training stage 3.
- **Strategy 3.** The loss constraint is imposed only on the 2D degradation representation during training stage 3.

TABLE V
ABLATION STUDY ON THE TRAINING STRATEGY OF DEGRADATION GUIDANCE

Strategies	$\mathcal{L}_{d1}$	$\mathcal{L}_{d2}$	Stage 2	PSNR↑	SSIM↑	LPIPS↓
Strategy 1	✗	✗	✗	29.82	0.8607	0.1967
Strategy 2	✓	✗	✗	29.98	0.8635	0.2036
Strategy 3	✗	✓	✗	29.97	0.8617	0.1991
Strategy 4	✓	✓	✓	30.15	0.8654	0.2038
Ours	✓	✓	✗	**30.24**	**0.8682**	**0.1941**

TABLE VI
ABLATION STUDY ON THE TRAINING STRATEGY OF SEMANTIC RESTORATION

Strategies	Stage 2	Stage 3	PSNR↑	SSIM↑	LPIPS↓
Strategy A	✗	✗	30.07	0.8643	0.2011
Strategy B	✓	✗	29.98	0.8635	0.2080
Strategy C	✗	✓	30.07	0.8657	0.2016
Ours	✓	✓	**30.24**	**0.8682**	**0.1941**

- **Strategy 4.** Loss constraints are imposed on both types of degradation representation during training stage 3, and the student MoEDE is first trained to convergence in training stage 2 before joint optimization.
- **Ours.** The default training strategy.

The experimental results are reported in Table V. As can be observed, the teacher MoEDE generates both types of degradation representation by assessing the difference between rainy image and ground-truth, so its output can provide a reference for the student MoEDE. Therefore, imposing the corresponding loss constraints is beneficial. Training the student MoEDE to convergence before joint optimization does not yield further performance gains.

3) Training Strategy of Semantic Restoration: To verify the validity of the proposed semantic restoration mechanism in training strategy, the following ablations were conducted:

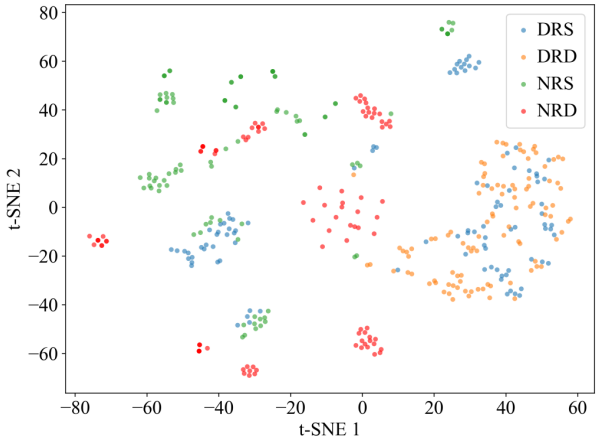
- **Strategy A.** DINOv3 is not retrained, and semantic compensation is performed using its pretrained weights.
- **Strategy B.** DINOv3 is trained only in stage 2 and kept frozen during joint optimization in stage 3.
- **Strategy C.** DINOv3 is optimized only during joint training in stage 3, without being trained to convergence in stage 2 beforehand.
- **Ours.** The default training strategy.

The experimental results are reported in Table VI. We can see that DINOv3 representations extracted from rainy images contain degradation-related interference, which prevents them from providing reliable semantic compensation. To ensure that semantic feature restoration benefits image deraining, DINOv3 must be optimized in a joint way with the deraining network. Furthermore, training DINOv3 to perform semantic restoration is challenging. Training DINOv3 to convergence in stage 2 substantially reduces the optimization burden during joint training in stage 3, thereby improving deraining performance.

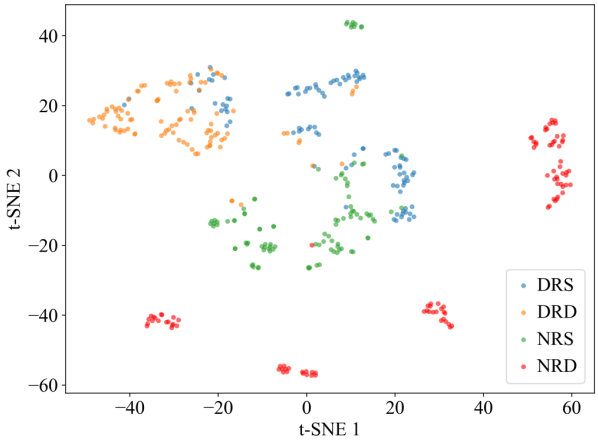
4) Dimension of the Degradation Representations: To analyze how the dimension of the degradation representations affects deraining performance, different values of L were evaluated. As shown in Table VII, the proposed model achieves optimal performance with $L = 96$. When L is too small, the MoEDE may lack sufficient capacity to depict degradation characteristics. Conversely, increasing L excessively may introduce a higher risk of overfitting.

TABLE VII
ABLATION STUDY ON THE DIMENSION OF THE DEGRADATION REPRESENTATIONS

Value of L	PSNR↑	SSIM↑	LPIPS↓
32	30.13	0.8655	0.1987
64	30.18	0.8661	0.1975
96	**30.24**	**0.8682**	**0.1941**
128	30.06	0.8650	0.1987



(a) 1D degradation representations of ground-truth images



(b) 1D degradation representations of rainy images

Fig. 11. T-SNE analysis of 1D degradation representations on the RainRAG test subsets [39].

D. Motivation Verification

1) Degradation Guidance: To verify that the 1D degradation representation can indeed reflect rain types, the trained student MoEDE was used to extract 1D degradation representations from images in the RainRAG test subsets [39]. The resulting representations were visualized using t-SNE.

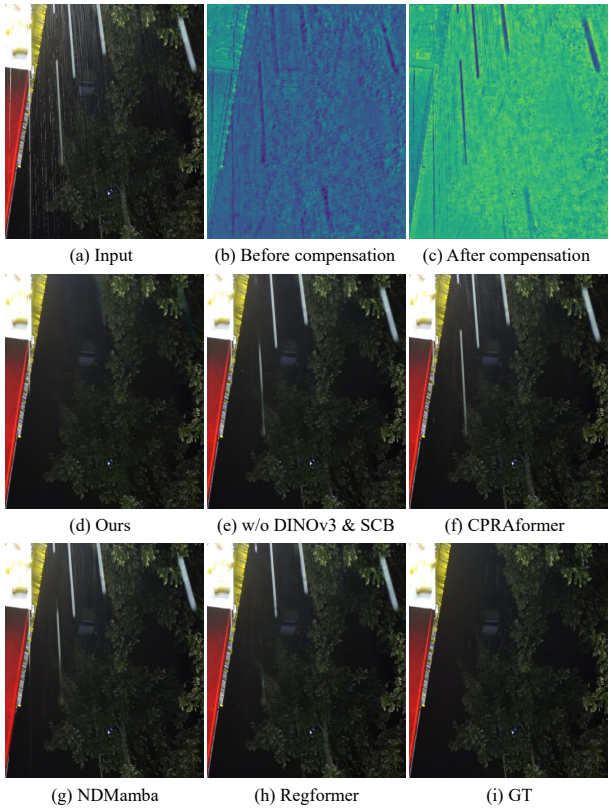


Fig. 12. A visualization of the semantic restoration mechanism.

As shown in Fig. 11(a), the ground-truth images are free of rain, so the MoEDE can not infer rain types from them. Consequently, samples from different test subsets will not form clearly separated clusters, with evident overlap between DRS and DRD. In contrast, as shown in Fig. 11(b), once the rainy images are used as input, the MoEDE will be able to perceive the rain types, resulting in clearly separated clusters.

The mapping between the 2D degradation representation and the spatial distribution of rain is learned from data. Consequently, the visual appearance of the 2D representation does not directly reveal the spatial distribution of rain in an intuitively interpretable manner. Therefore, it is not visualized here and the effectiveness of 2D degradation representation has been validated through the ablation study.

2) Semantic Restoration: To prove that the proposed semantic restoration mechanism can truly facilitate rain distinguishing, we chose a representative image for decoder feature visualization. As shown in Fig. 12, when rain streaks in an image exhibit substantial scale variations, the model may mistake some rain streaks as background components and fail to remove them. Before semantic compensation, the feature responses in rain streak regions are not sufficiently distinct from those in background regions. After semantic compensation, the contrast between two regions increases substantially. Fig. 12(e) shows the deraining result obtained by removing DINOv3 and SCB. The deraining results produced by the other three competitive methods are presented in Fig. 12(f) to Fig. 12(h). As can be observed, only the proposed model successfully handles this challenging case.

V. Conclusion

In this work, a unified image deraining framework is developed by coupling degradation-aware restoration with semantic compensation. The degradation representations characterize complementary aspects of rain and regulate state-space computing, while clean semantic representations recovered from rainy inputs provide additional cues for rain distinguishing and background preservation. The staged optimization strategy connects reference-guided learning with rainy-image-only inference. Ablation studies demonstrate the complementary contributions of degradation guidance and semantic restoration to the overall restoration process. Furthermore, evaluations on external datasets and real-world rainy photographs indicate that the learned deraining capability generalizes well beyond the training benchmark. These findings support the effectiveness of integrating degradation guidance and semantic restoration within a unified image deraining framework.

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